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FAULT DETECTION AND DIAGNOSIS IN ENGINEERING SYSTEMS

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Group project

TABLE OF CONTENTS

TABLE OF CONTENTS

ABBREVIATION

1	INTRODUCTION	1
2	LITERATURE REVIEW	2
2.1	Principal Component Analysis (PCA).....	2
2.2	Dynamic Principal Component Analysis (DPCA)	2
2.3	Monitoring of multiphase flow facility	3
2.4	Cranfield Flow Facility.....	3
2.5	Methods	4
2.5.1	PCA	4
2.5.2	DPCA	5
3	RESULTS	7
3.1	Fault Case 3.1 – Top Separator Input Blockage	7
3.2	Fault Case 4.3 – Bypass Valve Open	8
3.3	Discussion.....	10
4	CONCLUSIONS	12

REFERENCES

ABBREVIATION

Abbreviation	Full Term	Meaning / Description
FDD	Fault Detection and Diagnosis	A process used to monitor, detect, and diagnose faults in engineering systems.
PCA	Principal Component Analysis	A statistical technique for dimensionality reduction and fault detection.
DPCA	Dynamic Principal Component Analysis	An extension of PCA that includes time-lagged variables for dynamic systems.
CVA	Canonical Variate Analysis	A multivariate statistical method used for system monitoring under varying conditions.
PLS	Partial Least Squares	A statistical method used for modeling relations between input and output variables.
DPLS	Dynamic Partial Least Squares	Time-dependent extension of PLS for process monitoring.
SVD	Singular Value Decomposition	A method used in PCA and DPCA to decompose data into principal components.
KDE	Kernel Density Estimation	A non-parametric way to estimate the probability density function of a random variable.

1 INTRODUCTION

Fault Detection and Diagnosis (FDD) is a critical aspect of maintaining the reliability and efficiency of industrial processes. As industries evolve towards more flexible manufacturing schemes, the ability to monitor and adapt to changing operational conditions becomes increasingly important. This project focuses on the application of FDD methods to the Cranfield Multiphase Flow Facility, a complex system designed to simulate real-world industrial processes involving multiphase flow.

The Cranfield Multiphase Flow Facility, detailed in the paper by C. Ruiz-Cárcel et al., serves as an ideal benchmark for evaluating various fault detection techniques. The facility's ability to operate under varying conditions and its intricate setup provide a robust test bed for assessing the performance of different FDD methodologies. The data collected from this facility offers valuable insights into the behavior of multiphase flow systems and the challenges associated with detecting and diagnosing faults in such environments.

In this project, we aim to implement and compare at least two fault detection methods on the Cranfield Multiphase Flow Facility. By analyzing the provided data and applying advanced statistical techniques, we seek to identify and diagnose faults that may occur during the operation of the facility. Our goal is to demonstrate the effectiveness of these methods in real-world scenarios and provide recommendations for improving fault detection in industrial processes.

The project will involve a thorough understanding of the case study, data analysis, implementation of fault detection algorithms, and validation of results. We will leverage the knowledge gained from lectures and explore additional methodologies to ensure a comprehensive approach to fault detection and diagnosis. The findings from this project will contribute to the ongoing efforts to enhance the reliability and efficiency of industrial systems through advanced monitoring techniques.

2 LITERATURE REVIEW

Fault Detection and Diagnosis (FDD) is an essential aspect of maintaining reliability of industrial equipment. The ability to detect and diagnose faults early can prevent costly downtime, reduce maintenance costs, and ensure the safety of operations. This chapter provides an overview of the theoretical background of FDD, focusing on the principles and methodologies used later in the project.

2.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a statistical technique used for dimensionality reduction while retaining as much variability as possible within the dataset. It transforms correlated variables into a new set of uncorrelated principal components, which are orthogonal and ranked by the variance they capture. This makes PCA particularly useful for fault detection and diagnosis in complex systems, where analyzing multiple variables simultaneously can reveal underlying issues efficiently (Mansouri & ZS, 2015; Mansouri et al., 2016).

The process begins with normalizing the data to ensure equal contribution from all variables. The covariance matrix is then computed to identify relationships between variables. Singular Value Decomposition (SVD) is applied to extract the principal components, which indicate the directions of maximum variance. By projecting the data onto these components, its dimensionality is reduced while preserving critical information. In fault detection, deviations in the principal components can highlight anomalies, enabling early identification of potential system failures.

2.2 Dynamic Principal Component Analysis (DPCA)

Dynamic Principal Component Analysis (DPCA) is an extension of the PCA that adds the ability to understand changes over time instead of just the singular output. This is done by extending the data to time-lagged measurements, so instead of only feeding data at the current time, additional data from previous times are also considered. DPCA is typically

effective at detecting time delays and dynamic faults that PCA struggles with. Theoretically DPCA has an edge over PCA in the Cranfield Multiphase Flow Facility in a couple fields, as many of the sensors—such as flow meters— are time dependent, that identify blockages or changes in fluid density. However, it also contains a lot of non-time dependent sensors where PCA has an edge.

2.3 Monitoring of multiphase flow facility

This part of the project is focused on watching over and monitoring a system called the Cranfield Multiphase Flow Facility. This System deals with the movement of two or more different substances. The aim of this project is to test two different methods of fault detection for Cranfield Multiphase Flow Facility.

2.4 Cranfield Flow Facility

The case study is an extensive investigation into applying statistical process monitoring techniques on an actual industrial-sized system, a three-phase multiphase flow facility at Cranfield University. The facility is set up to mimic a real industrial process with controlled and monitored flows of water, oil, and air through pipelines, separators, valves, and sensors.

The key aim of the research was to create a benchmark case study with true experimental data to compare and analyze the performance of different fault detection process monitoring methods. Among them, Canonical Variate Analysis (CVA) was

the prime interest because it is renowned for its reliability in coping with systems at varying operating conditions. The other popular methods like Principal Component Analysis (PCA), Dynamic PCA (DPCA), Partial Least Squares (PLS), and Dynamic PLS (DPLS) were also covered for comparative purposes.

Six fault models of different types were intentionally introduced into the system to simulate real-life faults that may or will take place in real industrial operations. These fault models enabled a check on how the different fault detection techniques would respond to them when introduced under stable and fluctuating flow conditions. The fault

models introduced involved clogging in the water and air supply lines, clogging at the top separator inlet, opening a bypass line to simulate leakage, slugging phenomenon, and pressurization of an isolated pipe. The process data from 24 variables, comprising valve positions, temperatures, pressures, and flow rates, were studied. Canonical Variate Analysis was implemented on this data to track system behavior as well as identify fault initiation.

The findings illustrated that CVA, in conjunction with KDE, performed much better than other techniques with regards to fault detection accuracy and fault-tolerance to varying conditions. The research therefore concludes that CVA is an exceptionally powerful method for real-time monitoring and fault diagnosis in industrial systems, and the benchmark case used here presents a rich source for future research and development.

2.5 Methods

The two techniques chosen for this project are Principal Component Analysis (PCA) and Dynamic Principal Component Analysis (DPCA). Although Canonical Variate Analysis (CVA) has been demonstrated to provide better performance in other studies, PCA and DPCA were chosen because we were exposed to them extensively in the course lectures, and as such, had a better appreciation for how they work. This gave us a much better handle on how to implement corresponding MATLAB code. In addition, comparing PCA and DPCA provides considerable insight into how including time-lagged variables in DPCA improves its sensitivity to system dynamics, thereby enabling an examination of the dimensional benefits that can result from dynamic modelling.

2.5.1 PCA

The MATLAB code for the PCA fault detection consists of two phases. The first phase is the training phase, in which the parameters for PCA fault diagnosis are determined based on fault-free data. In the second phase, the test phase, these parameters are used to analyse the existing data for anomalies.

The training phase begins by loading the existing datasets with fault-free data and combining them in a single matrix. The dataset must be normalised so that the parameters

created from the dataset can be used for all other data. This means that the parameters can be applied to all data regardless of the scaling. This normalised matrix is then decomposed by singular value decomposition into the eigenvectors and diagonal matrix from these to identify the maximum variance of the data. To get a better performance of the programme we decided to reduce the dimension of the data to 3. In this way, the three most important feature combinations are considered, which contain a large part of the data. Based on this normalised and reduced data, the T^2 statics and the Q statics value are now calculated. As we have only used fault-free data as training data, the threshold is set to the maximum value. This means that all data with a greater deviation than the training data will later be identified as anomalies. The last step in the training phase is that we plot these thresholds for visualisation.

Then we start the test phase. At the beginning, this is very similar to the training phase. The data set that which must be checked for faults is loaded. This is then normalised in the same way as the training data and reduced to the same number of dimensions. The T^2 static and Q static values are then calculated again using this reduced and normalised data. These values are now compared with the threshold calculated from the training data. All values that exceed this threshold are recognised as faults. For the visual representation, the two values T^2 and Q are displayed in a plot and all values that exceed the threshold are highlighted in grey.

Since multiple datasets must be checked, we have packed the entire process into a function, that we only must call it with the new data record which eliminates the need to rewrite the entire code each time.

2.5.2 DPCA

The MATLAB code for fault detection with DPCA also consists of the same two main phases: the training phase and the test phase.

In the training phase, the fault-free data set is loaded and merged into a matrix. In contrast to the PCA approach, the DPCA approach not only considers the current values, but also their time delay. For this purpose, a so-called lagged matrix is created, which also contains the values of the previous points in time for each value. This makes it possible to recognise dynamic dependencies during the process. This combined matrix of current and previous values is then normalised as in the PCA approach and converted into the

eigenvectors using singular value decomposition. With the reduction of the dimension to three principal components we also get the data to calculate the T^2 and Q values. Since only fault-free data is used during training, the thresholds for both T^2 and Q statistics are set to the maximum values observed in the training set. Any future observation exceeding these thresholds will be classified as a potential anomaly. Finally, T^2 and Q statistics are plotted along with their thresholds for visual inspection.

In the test phase, the trained model is used to detect faults in new datasets. For each dataset, the same steps as in the training phase are applied: the creation of the lagged matrix, normalization using the training data's parameters, projection onto the principal components, and computation of the T^2 and Q statistics.

These statistics are then compared to the thresholds calculated during training. Any values that exceed the thresholds are flagged as potential faults. In the plots, these fault points are visually highlighted, allowing for clear identification of anomalies.

To improve code reusability and minimize repetition, the testing procedure is encapsulated in a function. This function takes a new dataset and automatically performs all necessary steps making the fault detection process efficient and scalable across multiple datasets.

3 RESULTS

This chapter presents the results obtained during the work. In the thesis instructions document, you will find instructions on the use and formatting of images, graphs, tables and formulas.

The results of applying PCA and DPCA to Fault Case 3.1 and Fault Case 4.3 in the Cranfield Multiphase Flow Facility dataset will be presented as an example. The goal is to assess the ability of these multivariate statistical techniques to detect faults under both steady-state and dynamic operating conditions. The analysis uses both T^2 and Q -statistics to assess the ability of each method to detect and isolate faults.

3.1 Fault Case 3.1 – Top Separator Input Blockage

Figure 1 shows the Q -statistic for Fault Case 3.1. The PCA and DPCA curves exhibit identical shapes, with the DPCA plot slightly shifted upward. From sample 600 to 1400, the values fluctuate around the threshold, with most points lying above it. This indicates that both methods detect the fault in this time window, and the vertical offset in DPCA does not yield a functional improvement. Thus, the Q -statistic offers comparable fault sensitivity for PCA and DPCA in this scenario.

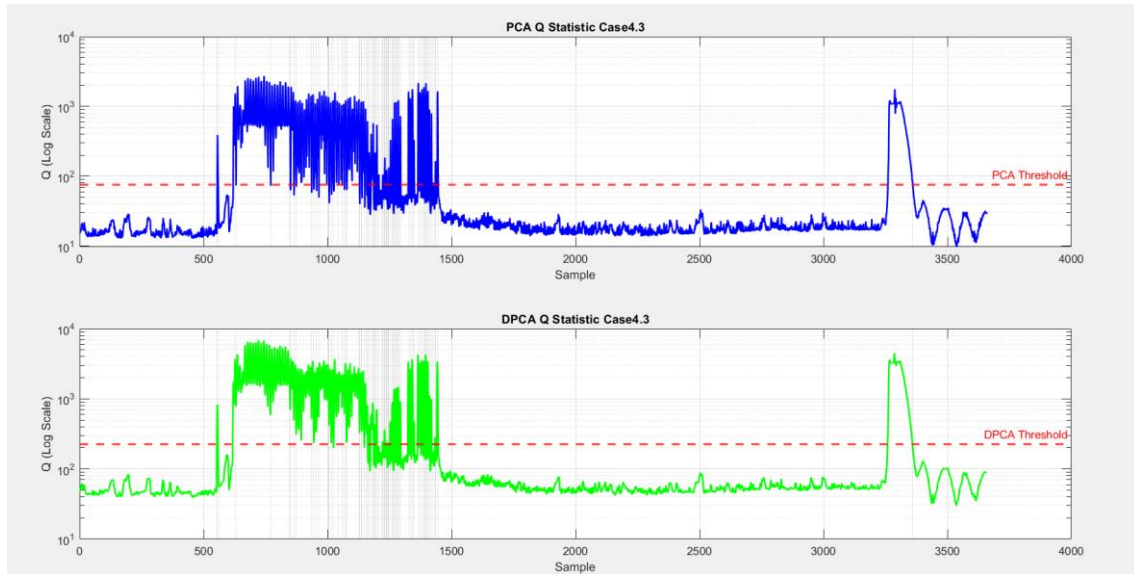


Figure 1 Q -statistic Fault Case 3.1

Figure 2 presents the T^2 -statistic for the same fault case. Both PCA and DPCA curves are visually identical, with clear threshold crossings between samples 600 and 1400. This indicates that both methods respond equally to the fault using T^2 -statistics, and no advantage is provided by DPCA in this instance.

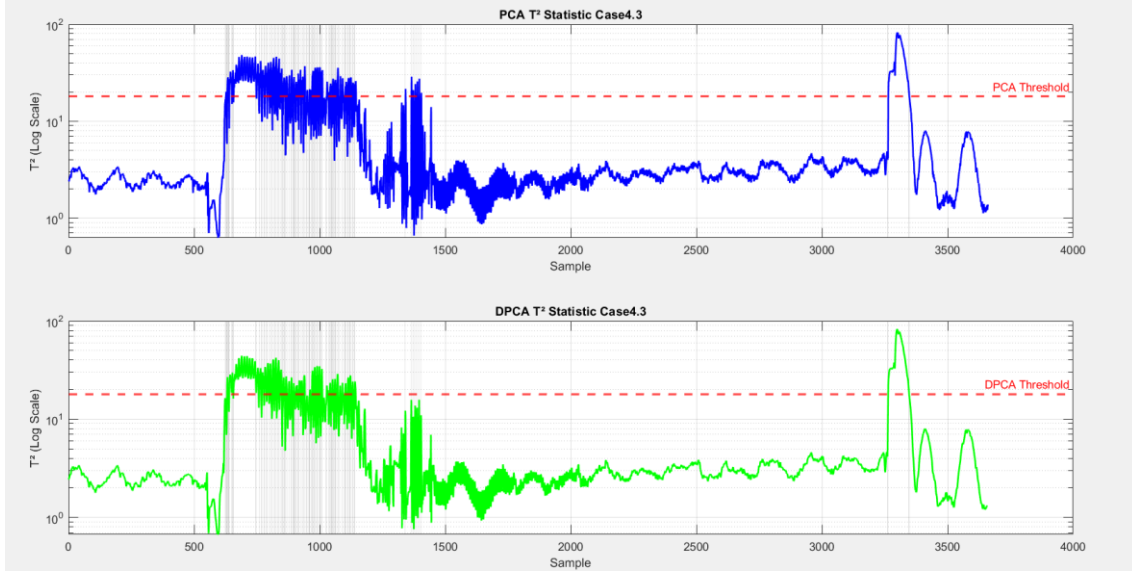


Figure 2 T^2 -statistic for Case 3.1

3.2 Fault Case 4.3 – Bypass Valve Open

Figure 3 shows the Q-statistic for Fault Case 4.3. Both PCA and DPCA display nearly identical curve shapes, with DPCA values shifted slightly higher across the full sample range. In both methods, the Q-statistic remains above the threshold from sample 1200 to 8400, suggesting that the fault is sustained and detectable by either technique. However, the shape similarity indicates no added value from DPCA's dynamic modeling in this

particular

case.

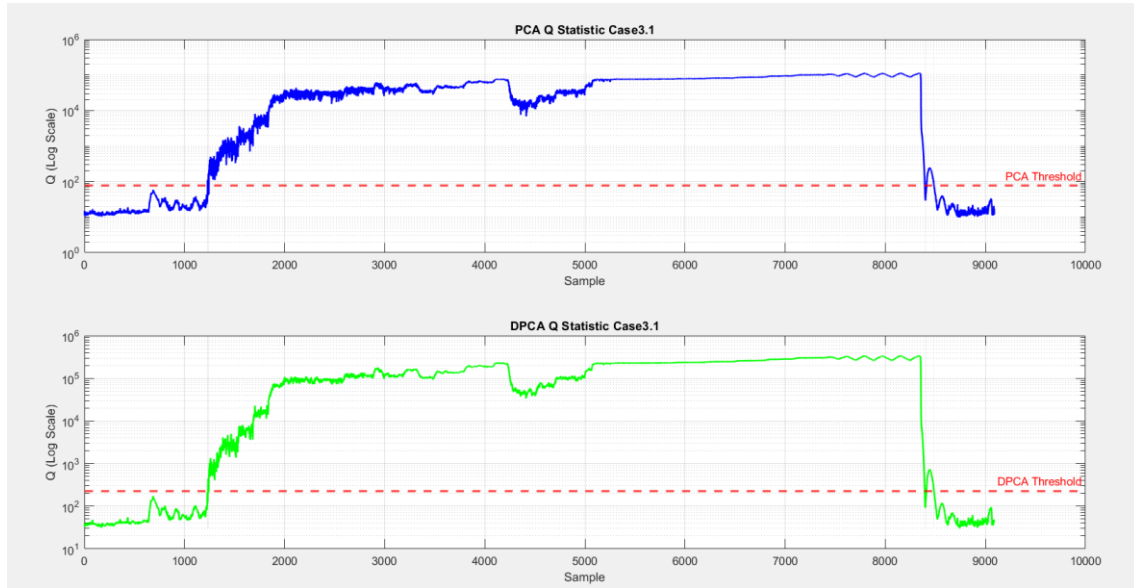


Figure 3 Q -statistic Fault Case 4.3

Figure 4 presents the T^2 -statistic for the same fault. As with the Q -statistic, the curves for PCA and DPCA are nearly identical, with both methods exceeding the threshold consistently between sample 1200 and 8400. This indicates that both PCA and DPCA are equally capable of detecting this fault, with no significant performance difference evident from this plot.

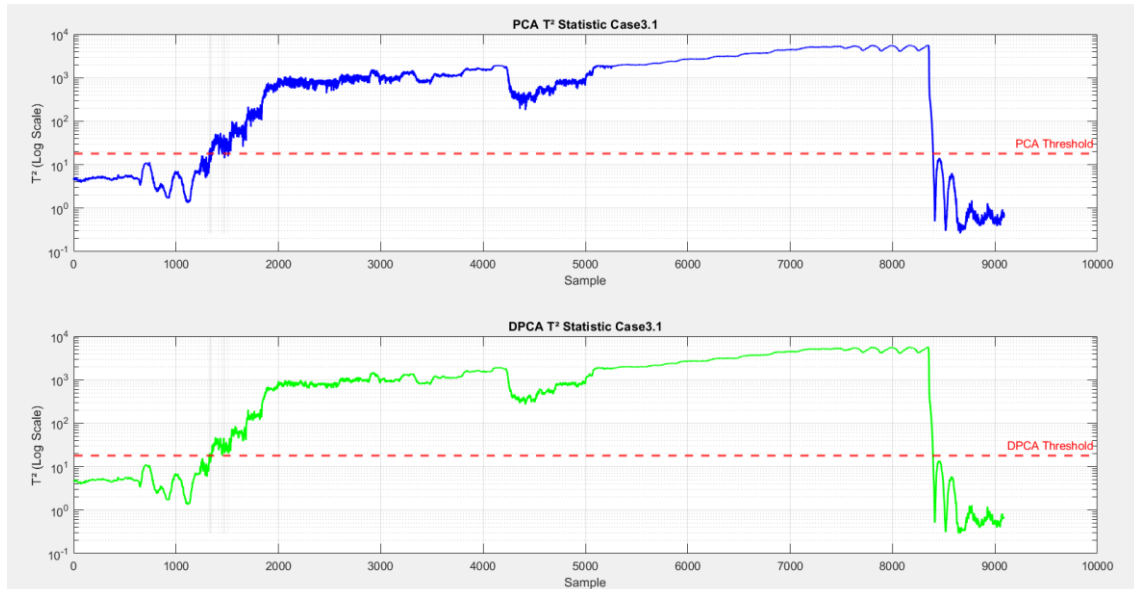


Figure 4 T^2 -statistic for Fault Case 4.3

The results show that DPCA can offer improvements over PCA in some fault scenarios, especially when temporal dynamics are relevant. However, in the specific cases analyzed (Fault Case 3.1 and 4.3), PCA and DPCA exhibited similar performance, both successfully detecting faults. DPCA's use of lagged variables can improve detection in more dynamic cases, though its advantage is not always pronounced.

3.3 Discussion

In this project, Principal component Analysis (PCA) and dynamic principal component analysis (DPCA) were the chosen methods for fault diagnosis in process monitoring. Even though Canonical Variate Analysis (CVA) was proven to be the best approach in the provided research for these varying conditions, DPCA and PCA were selected for our project due them being the most familiar ones. This familiarity orchestrated our successful implementation of these methods in MATLAB.

Our results confirmed that PCA is a reliable method for fault detection, although it works more like a static tool. It reduces the number of variables while keeping the main variation, but it doesn't really take time-based changes into account. Because of that, PCA worked well when the conditions were stable, but it sometimes missed faults that developed slowly or under changing conditions.

DPCA, on the other hand, includes time-related information by using lagged variables. This helped it react more quickly to system changes. In our case, DPCA gave better results than PCA in both how fast it detected problems and how accurate it was. Since it could catch faults earlier, it could reduce downtime and avoid bigger issues.

This is also in line with previous studies, such the one by Ruiz-Cárcel et al. (2015), which also found that DPCA performs better in systems that change over time. Even though DPCA is more complex and needs careful setup, especially when choosing lag values, its better performance makes it worth using in many situations.

Still, PCA is easier to use and understand, which is a big advantage, especially for simpler systems or when a quick check is needed. So, the choice between PCA and DPCA really depends on how dynamic the system is and what the project requires.

4 CONCLUSIONS

This chapter describes the conclusions drawn from the results, the analysis, and the recommendations for further research.

The project explored the options of PCA and DPCA for fault detection in the Cranfield multiphase flow facility. Both the PCA and DPCA detected the faults. However, it was expected that the DPCA would have an advantage with time-lagged variables, but the results showcase that they performed nearly identical.

In conclusion, both PCA and DPCA is viable for fault detection for our data. Furthermore, it should be highlighted that a more dynamic fault case would potentially highlight the difference between PCA and DPCA better. The future research of such a project, should include more dynamic fault cases, optimization of the lag selection, and include additional fault detection strategies.

REFERENCES

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